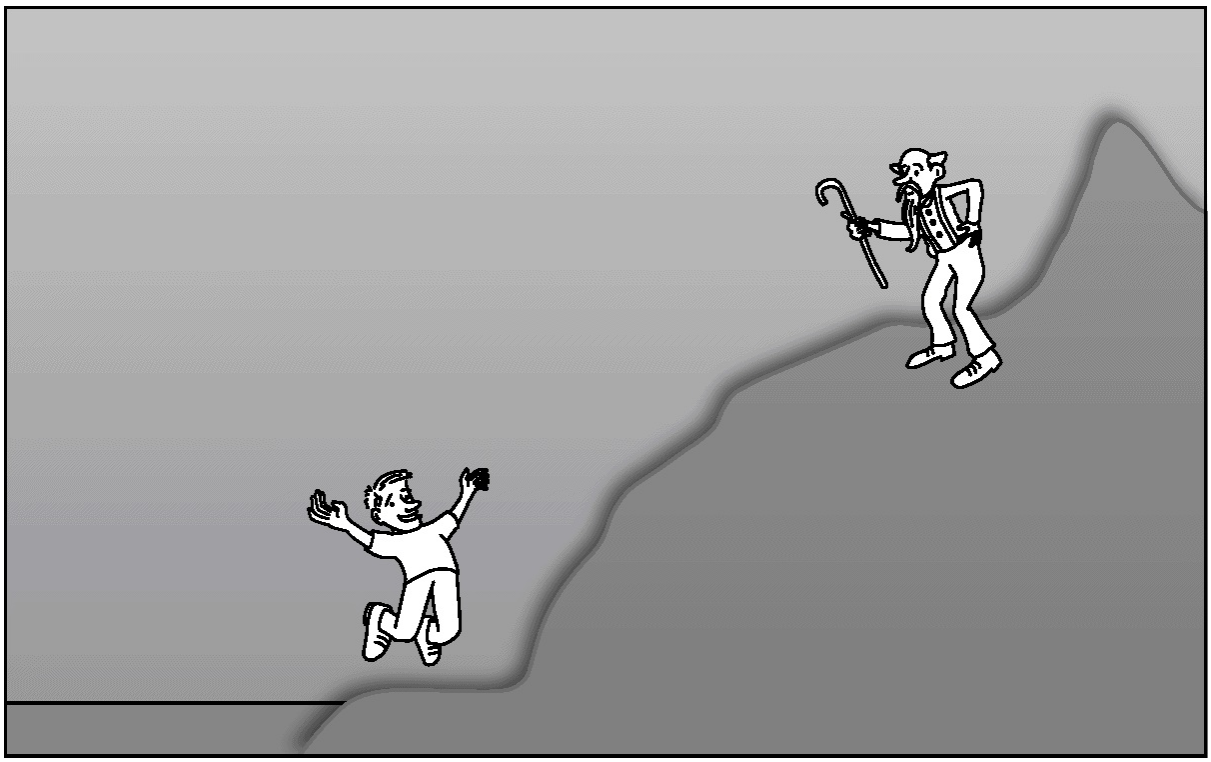


Figure 3.8 Two twins spend their time one at sea-level and the other in the mountains. When they meet up again, the twin who lived in the mountains is older. This is the gravitational dilation of time.

But the predictions of the theory go well beyond these minute effects. Stars burn as long as they have available hydrogen – their fuel – then die out. The remaining material is no longer supported by the pressure of the heat and collapses under its own weight. When this happens to a large enough star, the weight is so strong that matter is squashed down to an enormous degree and space curves so intensely as to plunge down into an actual hole. A black hole.